Computer Graphics
2D Transformations

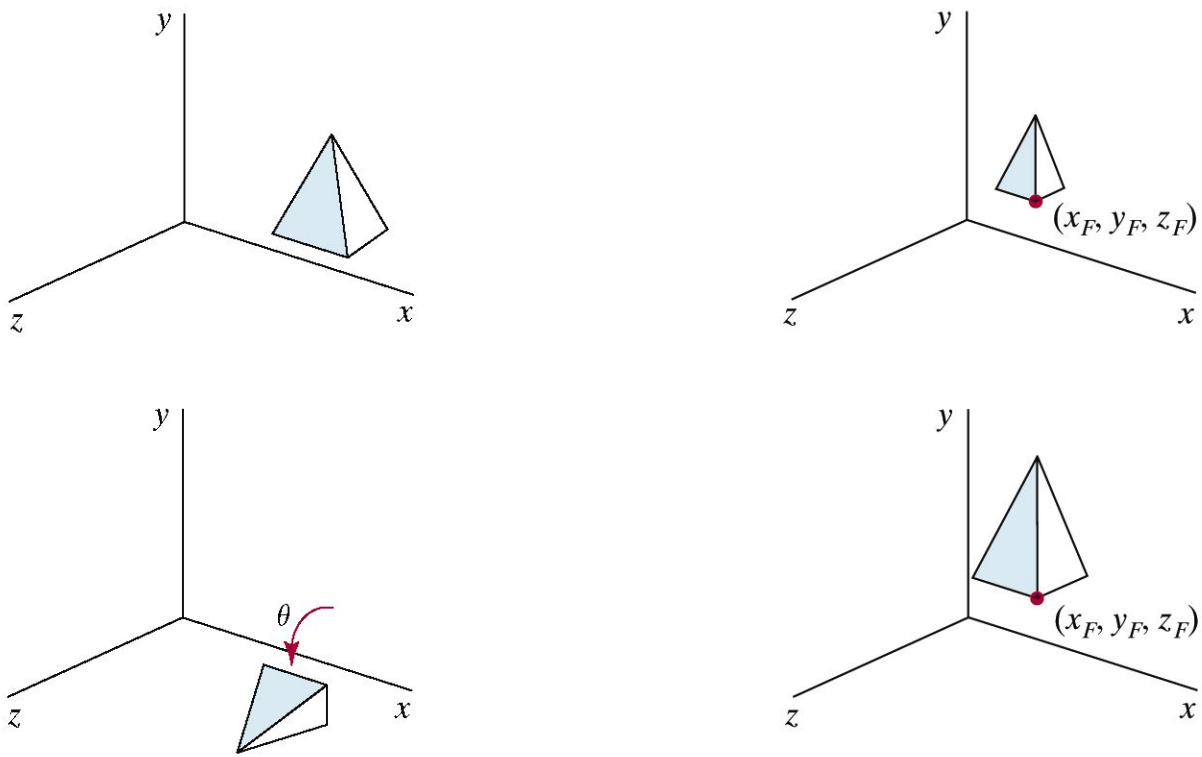
Contents

In today's lecture we'll cover the following:

- Why transformations
- Transformations
  - Translation
  - Scaling
  - Rotation
  - Shearing
  - Reflections
- Homogeneous coordinates
- Matrix multiplications
- Combining transformations

Why Transformations?

In graphics, once we have an object described, transformations are used to move that object, scale it and rotate it



Images taken from Hearn & Baker, "Computer Graphics with OpenGL" (2004)

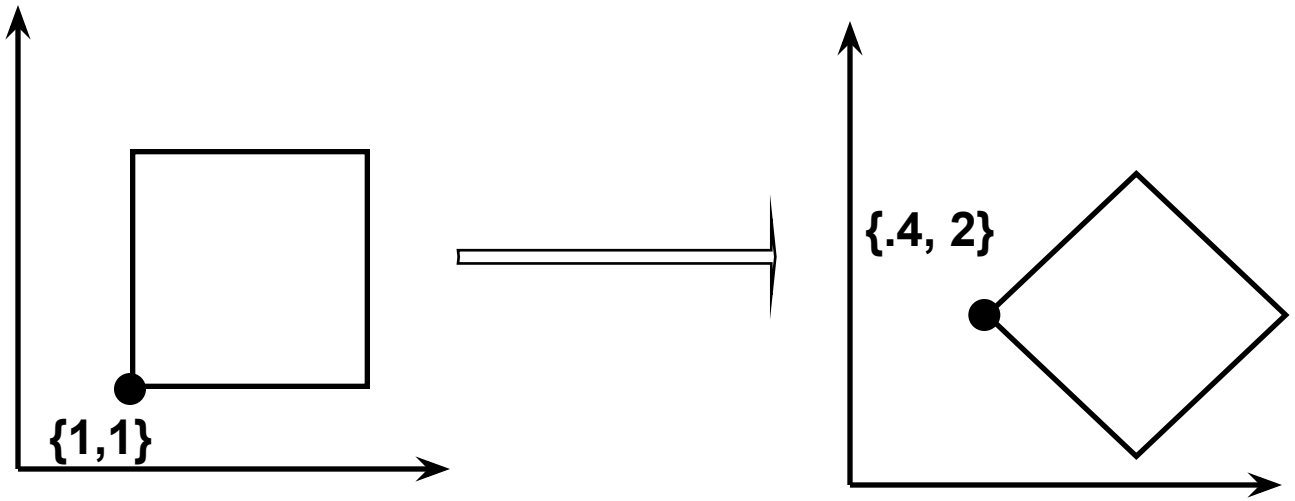
Transformations

A transformation on an object is an operation that changes how the object is finally drawn to screen

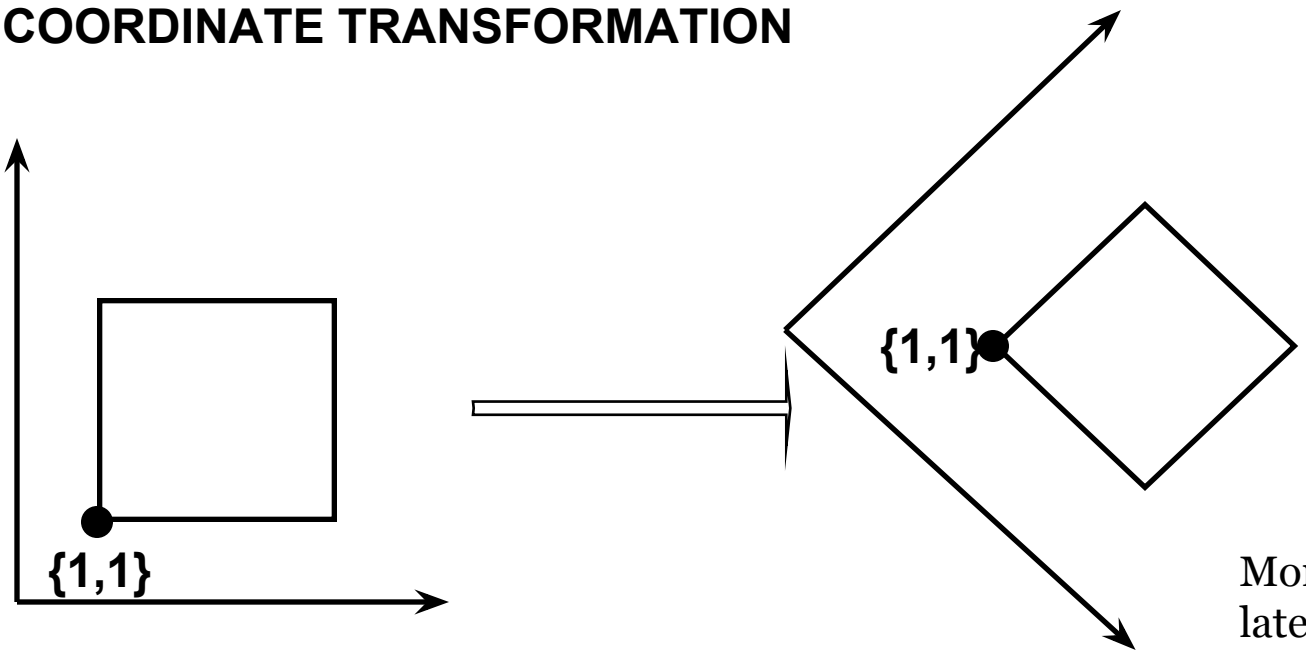
There are two ways of understanding a transformation

- An **Object Transformation** alters the coordinates of each point according to some rule, leaving the underlying coordinate system unchanged
- A **Coordinate Transformation** produces a different coordinate system, and then represents all original points in this new system

Example: OBJECT TRANSFORMATION



Example: COORDINATE TRANSFORMATION



More on this later...

Transformations

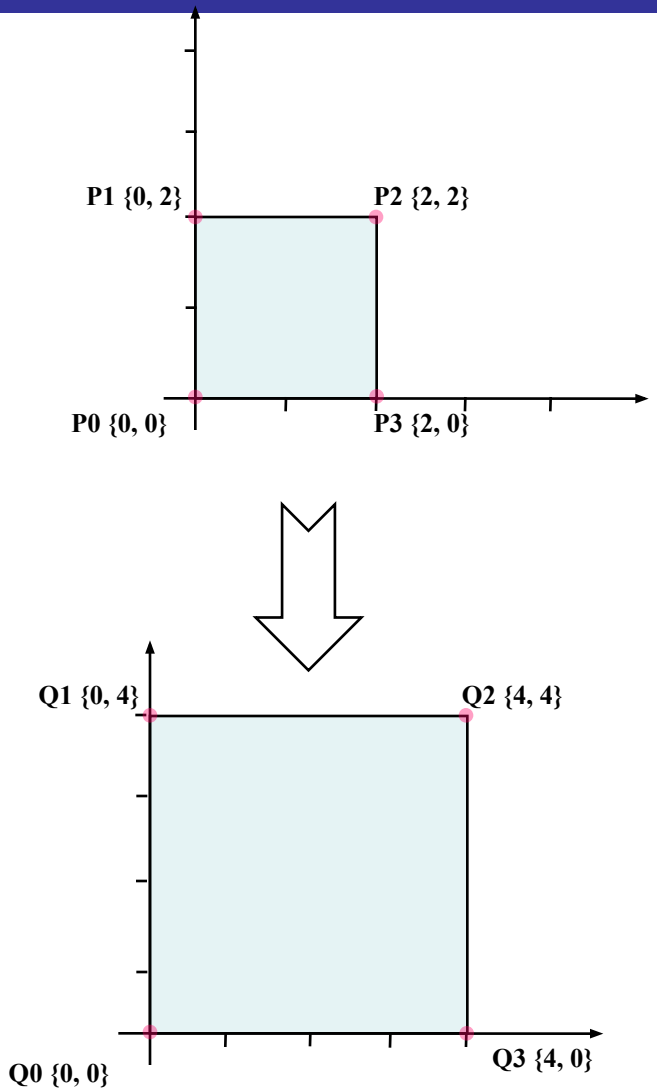
Whole collections of points may be transformed by the same transformation T, e.g. lines or circles

The image of a line, L, under T, is the set of all *images* of the individual points of L.

For most mappings of interest, this image is still a connected curve of some shape

For some mappings, the result of transforming a line may no longer be a line

Affine Transformations, however, do preserve lines, and are the most commonly-used transformations in computer graphics



Useful Properties of Affine Transformations

Preservation of lines:

Preservation of parallelism

Preservation of proportional distances

Why are they useful?

They preserve lines, so the image of a straight line is another straight line.

This vastly simplifies drawing transformed line segments.

We need only compute the image of the two endpoints of the original line and then draw a straight line between them

Preservation of co-linearity guarantees that polygons will transform into polygons

Preservation of parallelism guarantees that parallelograms will transform into parallelograms

Preservation of proportional distances means that mid-points of lines remain mid-points

Elementary Transformations

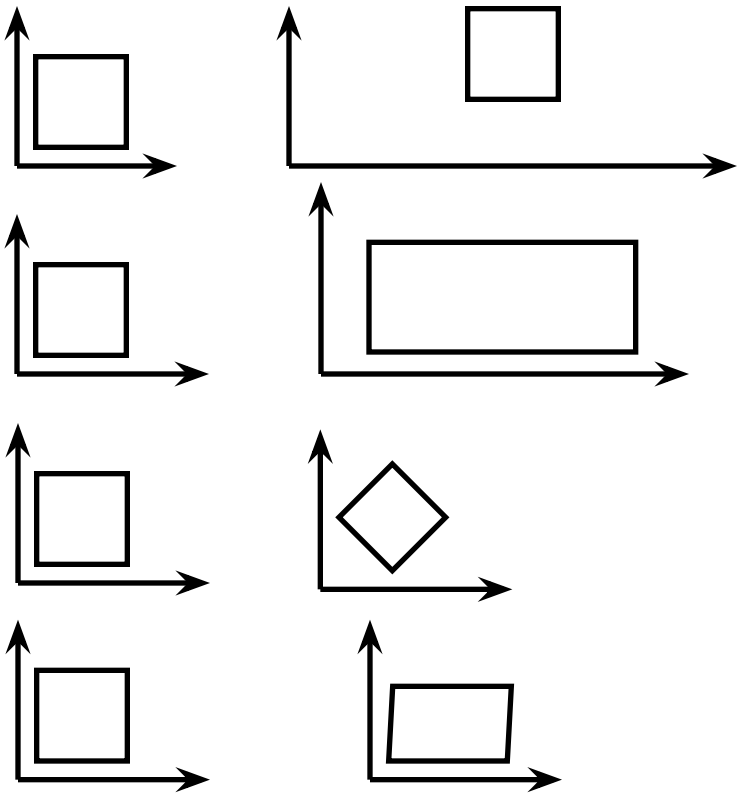
Affine transformations are usually combinations of four elementary transformations:

1: Translation

2: Scaling

3: Rotation

4: Shearing

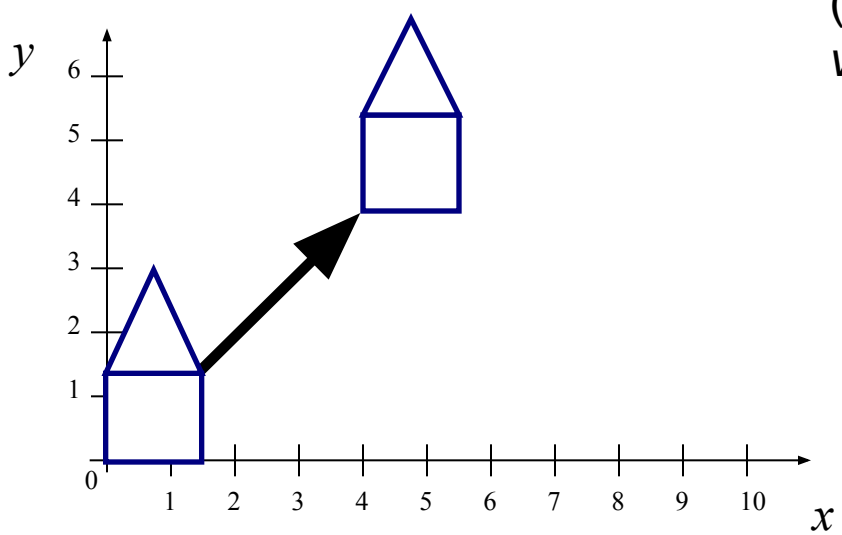


Translation

Simply moves an object from one position to another

$$x_{new} = x_{old} + tx \qquad y_{new} = y_{old} + ty$$

(tx,ty) is the translation vector or shift vector



Note: House shifts position relative to origin

Translation

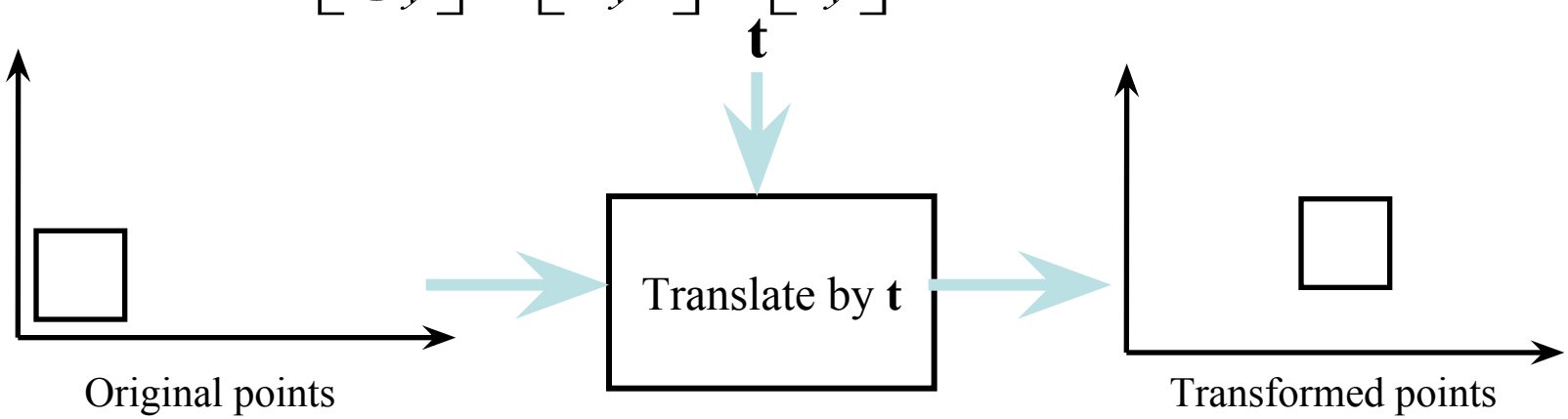
A **translation** moves an object into a different position in a scene

This is achieved by adding an offset/translation vector

$$\mathbf{t} = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

In Vector notation:

$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix} = \begin{bmatrix} P_x \\ P_y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$



Translation

$$P = \begin{bmatrix} x \\ y \end{bmatrix} \qquad P' = \begin{bmatrix} x' \\ y' \end{bmatrix} \qquad T = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

$$P' = P + T$$

Problem:

- Given a circle C with radius 10 and center coordinates (1, 4). Apply the translation with distance 5 towards X axis and 1 towards Y axis. Obtain the new coordinates of C without changing its radius.

Solution:

Given-

- Old center coordinates of C = (X_{old}, Y_{old}) = (1, 4)
- Translation vector = (T_x, T_y) = (5, 1)

- Let the new center coordinates of C = (X_{new}, Y_{new}).
- Applying the translation equations, we have-

$$X_{new} = X_{old} + T_x = 1 + 5 = 6$$
$$Y_{new} = Y_{old} + T_y = 4 + 1 = 5$$

- Thus, New center coordinates of C = (6, 5).

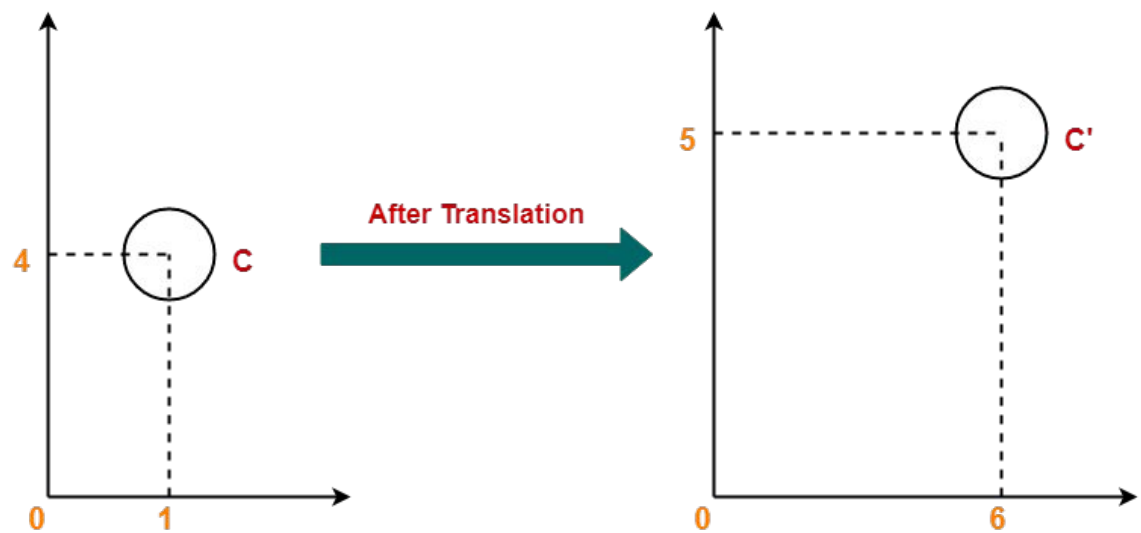
Alternatively,

In matrix form, the new center coordinates of C after translation may be obtained as-

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix} + \begin{bmatrix} T_x \\ T_y \end{bmatrix}$$

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} + \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$$



Exercise to do: Consider a square with coordinates A(0, 3), B(3, 3), C(3, 0), D(0, 0). Apply translation of 1 in X axis and 1 in Y axis.

Scaling

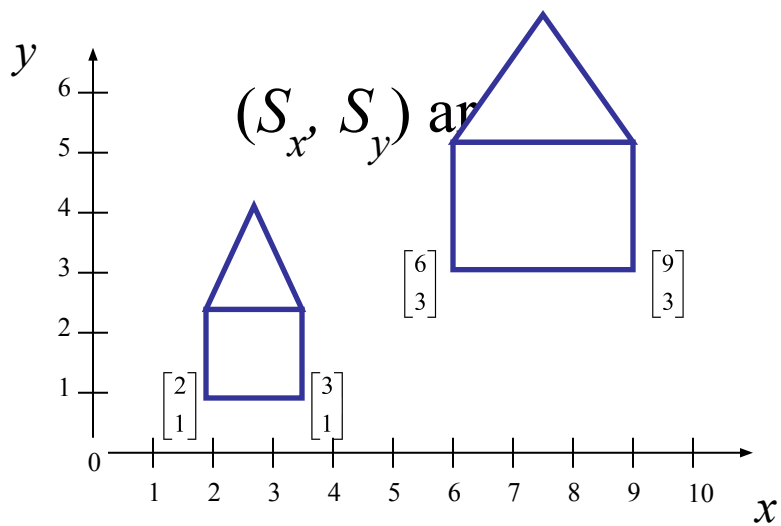
Scaling multiplies all coordinates

Alters the size of an object

WATCH OUT: Objects grow and move!

$$x_{new} = Sx \times x_{old} \qquad y_{new} = Sy \times y_{old}$$

(S_x, S_y) are the scaling factors



Note: House shifts position relative to origin

Scaling

$$P = \begin{bmatrix} x \\ y \end{bmatrix} \qquad P' = \begin{bmatrix} x' \\ y' \end{bmatrix} \qquad S = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

$P' = S.P$

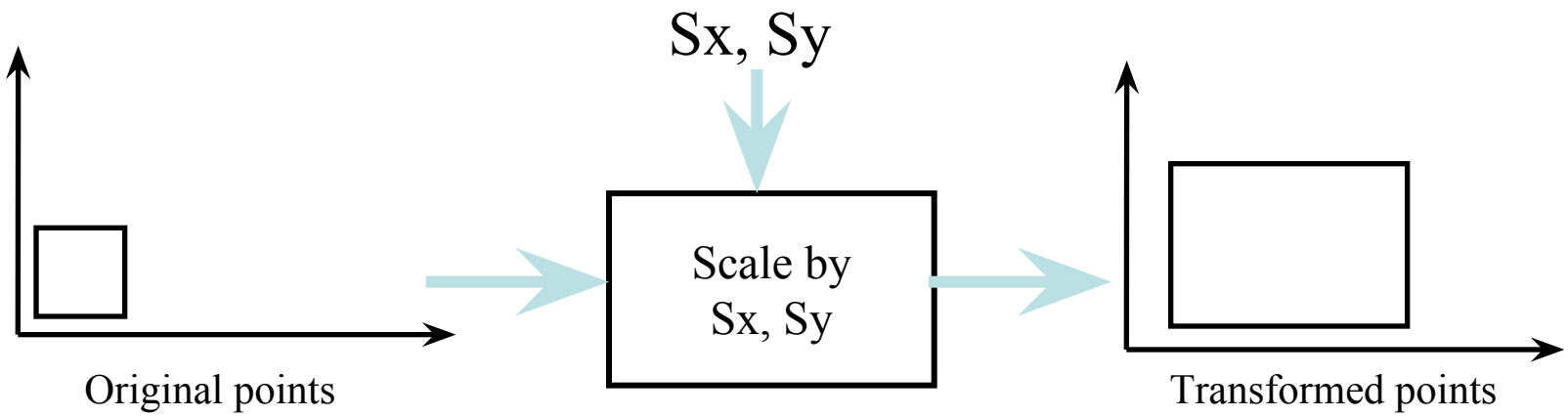
$S_x = S_y$ Uniform scaling

$S_x = S_y$ Differential scaling

Scaling

A scaling changes the size of an object with two scale factors, Sx and Sy (if Sx != Sy, then we have a non-uniform scale)

$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix} = \begin{bmatrix} P_x \times S_x \\ P_y \times S_y \end{bmatrix}$$



Fixed Point Scaling

- The location of a scaled object can be controlled by choosing a position, called the *fixed point*,(x_f, y_f) i.e., to remain unchanged after the scaling transformation.

- The fixed point can be chosen as the objects' centroid or any other position

- For a vertex with coordinates (x, y), the scaled coordinates (x', y') are calculated as,

$$x' = x_f + (x - x_f)s_x \qquad y' = y_f + (y - y_f)s_y$$

$$\Rightarrow x' = x.\ s_x + x_f\,(1 - s_x) \text{ and } y' = y.\ s_y + y_f\,(1 - s_y)$$

Where $x_f\,(1 - s_x)$ and $y_f\,(1 - s_y)$ are constants for all points in the object.

Problem: Given a square object with coordinate points A(0, 3), B(3, 3), C(3, 0), D(0, 0). Apply the scaling parameter 2 towards X axis and 3 towards Y axis and obtain the new coordinates of the object.

Solution: Given-

Old corner coordinates of the square = A (0, 3), B(3, 3), C(3, 0), D(0, 0)

Scaling factor along X axis = 2

Scaling factor along Y axis = 3

For Coordinates A(0, 3)

Let the new coordinates of corner A after scaling = (X_{new}, Y_{new}).

Applying the scaling equations, we have-

$X_{new} = X_{old} \times S_x = 0 \times 2 = 0$

$Y_{new} = Y_{old} \times S_y = 3 \times 3 = 9$

Thus, New coordinates of corner A after scaling = (0, 9).

For Coordinates B(3, 3)

Let the new coordinates of corner B after scaling = (X_{new}, Y_{new}).

Applying the scaling equations, we have-

$X_{new} = X_{old} \times S_x = 3 \times 2 = 6$

$Y_{new} = Y_{old} \times S_y = 3 \times 3 = 9$

Thus, New coordinates of corner B after scaling = (6, 9).

For Coordinates C(3, 0)

Let the new coordinates of corner C after scaling = (X_{new}, Y_{new}).

Applying the scaling equations, we have-

$X_{new} = X_{old} \times S_x = 3 \times 2 = 6$

$Y_{new} = Y_{old} \times S_y = 0 \times 3 = 0$

Thus, New coordinates of corner C after scaling = (6, 0).

For Coordinates D(0, 0)

Let the new coordinates of corner D after scaling = (X_{new}, Y_{new}).

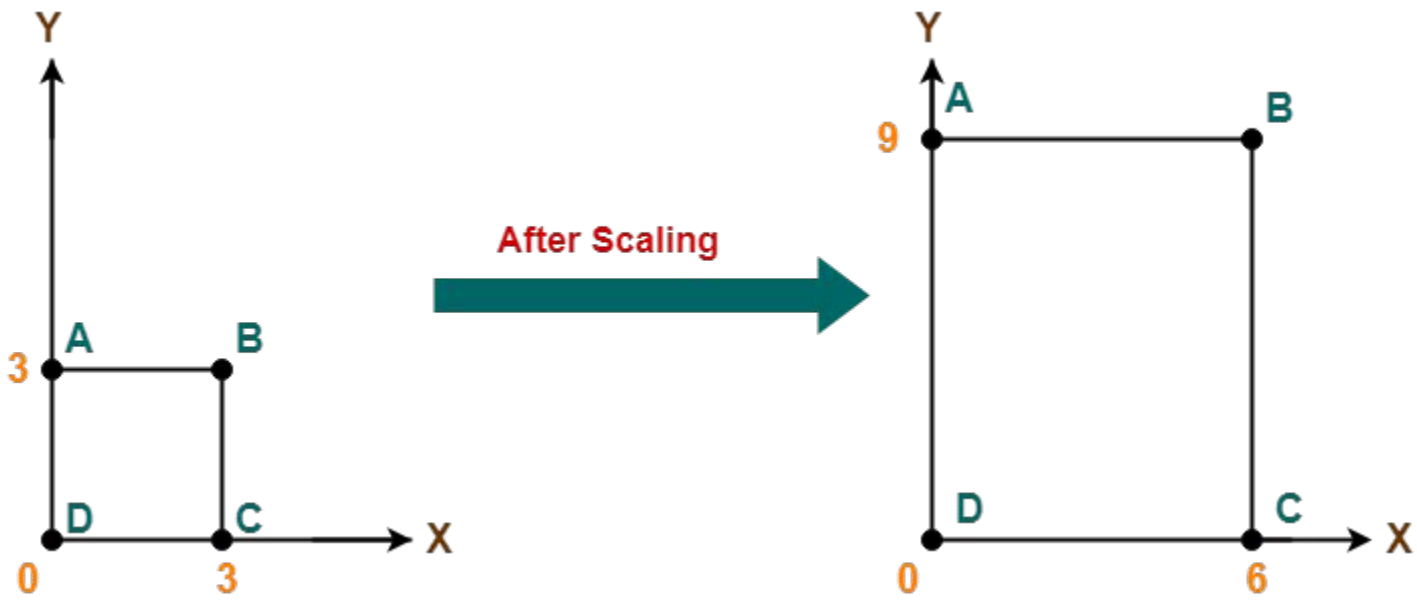
Applying the scaling equations, we have-

$X_{new} = X_{old} \times S_x = 0 \times 2 = 0$

$Y_{new} = Y_{old} \times S_y = 0 \times 3 = 0$

Thus, New coordinates of corner D after scaling = (0, 0).

Thus, New coordinates of the square after scaling = A (0, 9), B(6, 9), C(6, 0), D(0, 0).


Y
A
B
3
D
C
X
0
3
After Scaling
Y
A
B
9
D
C
X
0
6

Rotation

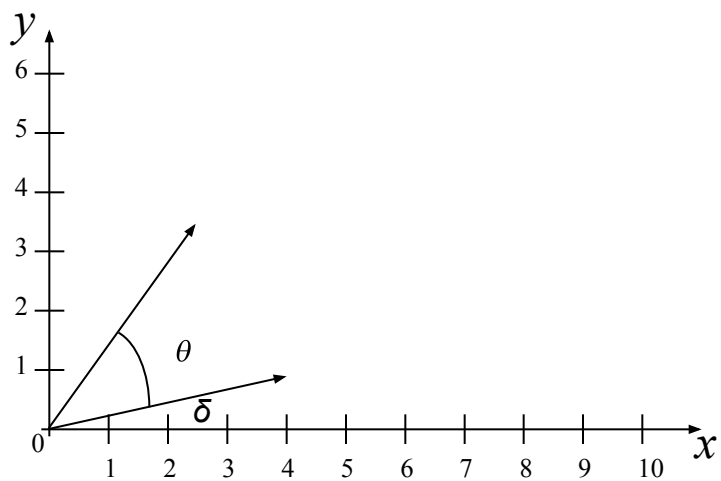
To generate a rotation, a rotation angle θ and the position (x_r, y_r) of the rotation point is needed

$$x_{new} = r \times \cos(\delta + \theta) = r \cos\delta \cos\theta - r \sin \delta \sin\theta$$

$$y_{new} = r \times \sin(\delta + \theta) = r \cos\delta \sin\theta + r \sin \delta \cos\theta$$

Original polar coordinates

$$x_{old} = r \cos \delta \text{ and } y_{old} = r \sin \delta$$



Rotation

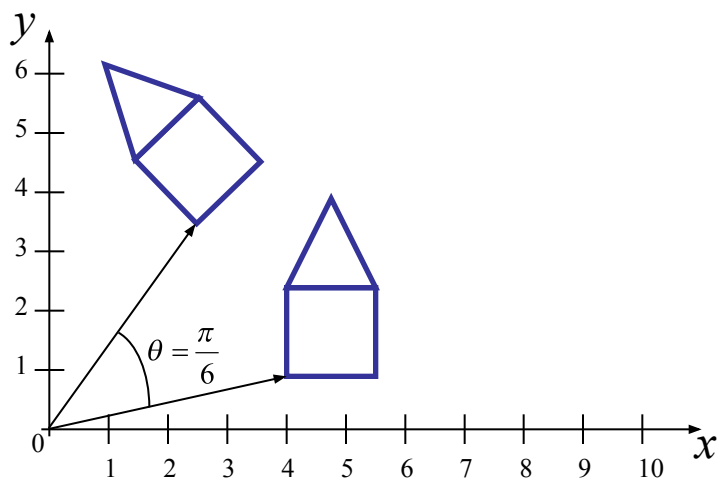
Rotates all coordinates by a specified angle

$$x_{new} = x_{old} \times \cos\theta - y_{old} \times \sin\theta$$
$$y_{new} = x_{old} \times \sin\theta + y_{old} \times \cos\theta$$

Points are always rotated about the origin

For rotation about any specified point (x_r, y_r):

$$x_{new} = x_r + (x_{old} - x_r) \times \cos\theta - (y_{old} - y_r) \times \sin\theta$$
$$y_{new} = y_r + (x_{old} - x_r) \times \sin\theta + (y_{old} - y_r) \times \cos\theta$$



Rotation equations can be written in the matrix form as

$P' = R.\ P$

Where rotation matrix

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

When coordinate positions are represented as row vectors instead of column vectors, the matrix product in rotation eqn (i) is transposed so that the trnsformed row coordinate [x' y'] is calculated as

$P'^{T} = (R.\ P)^{T} = P^{T}.R^{T}$

Where P^{T} = [x y], and the transpose matrix R^{T} of matrix R is obtained by interchanging rows & columns.

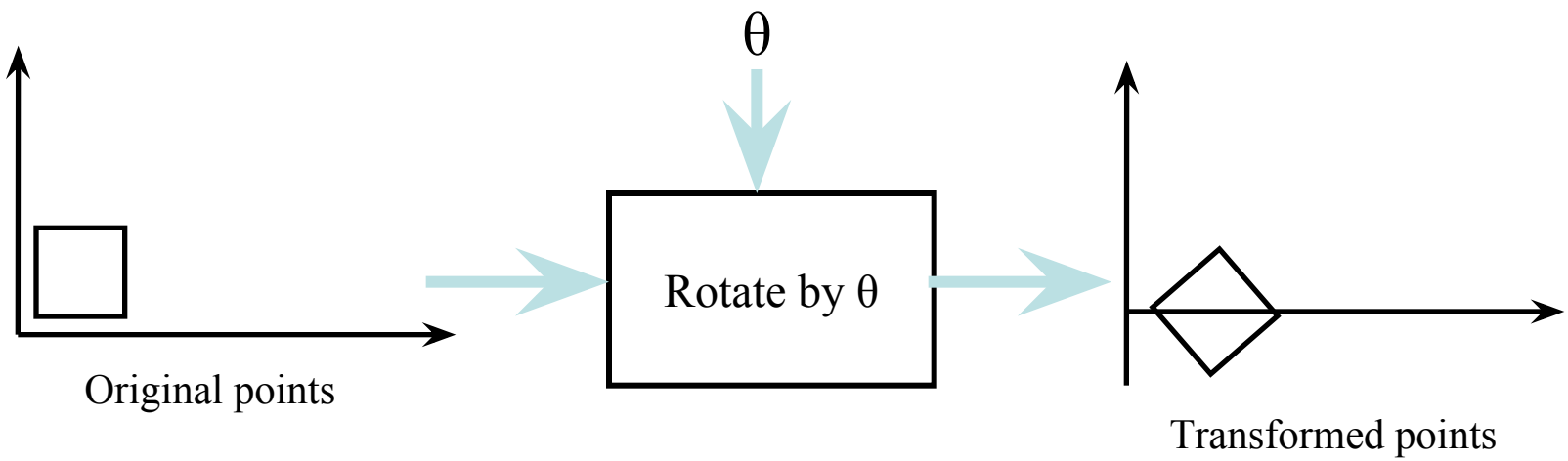
Rotation

Using the trigonometric relations, a point rotated by an angle q about the origin is given by the following equations:

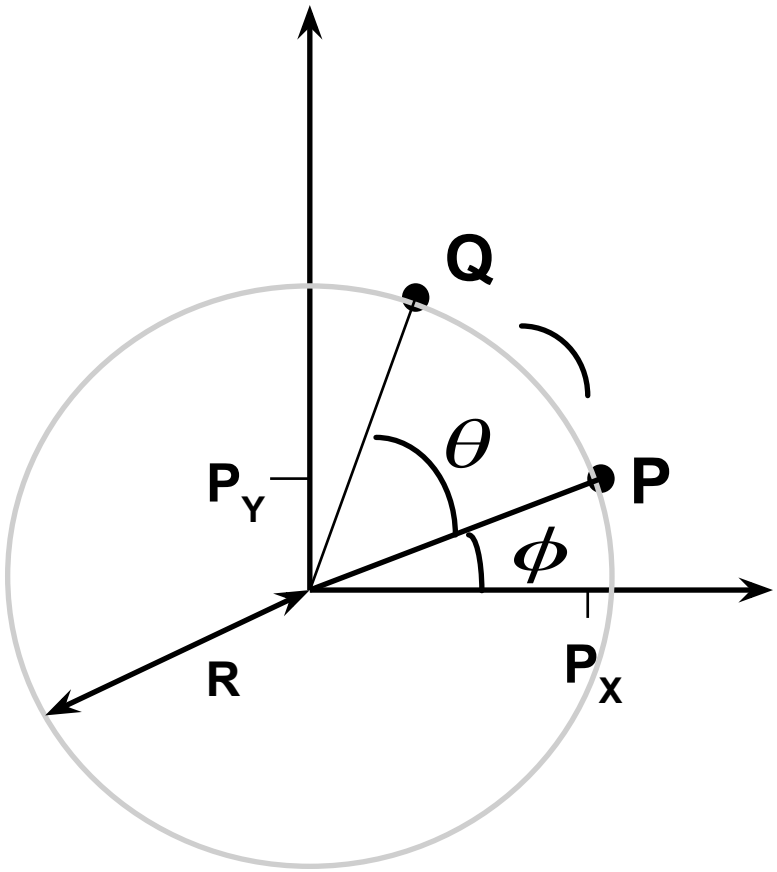
$$Q_y = P_x \sin\theta + P_y \cos\theta \qquad Q_x = P_x \cos\theta - P_y \sin\theta$$

So

$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix} = \begin{bmatrix} P_x \cos\theta - P_y \sin\theta \\ P_x \sin\theta + P_y \cos\theta \end{bmatrix}$$



Rotation - derivation



$$Q_X = R\cos(\theta + \phi) \qquad [1]$$
$$Q_y = R\sin(\theta + \phi) \qquad [2]$$

$$P_X = R\cos(\phi) \qquad [3]$$
$$P_y = R\sin(\phi) \qquad [4]$$

$$\boxed{\begin{aligned}\cos(\theta+\phi) &= \cos(\theta)\cos(\phi) - \sin(\theta)\sin(\phi)\\ \sin(\theta+\phi) &= \sin(\theta)\cos(\phi) + \cos(\theta)\sin(\phi)\end{aligned}}$$

[1] ⇨ $Q_x = R\cos(\theta)\cos(\phi) - R\sin(\theta)\sin(\phi)$

Substituting from [3] and [4]…

$$Q_x = P_x\cos(\theta) - P_y\sin(\theta)$$

Similarly from [2]…

$$Q_y = P_y\cos(\theta) + P_x\sin(\theta)$$

Rotation

Example: Given a triangle with corner coordinates (0, 0), (1, 0) and (1, 1). Rotate the triangle by 90 degree anticlockwise direction and find out the new coordinates.

Solution- We rotate a polygon by rotating each vertex of it with the same rotation angle.

Given-

- Old corner coordinates of the triangle = A (0, 0), B(1, 0), C(1, 1)
- Rotation angle = $\theta = 90^o$

For Coordinates A(0, 0)

Let the new coordinates of corner A after rotation = (X_{new}, Y_{new}).

Applying the rotation equations, we have-

X_{new}
$= X_{old} \times \cos\theta - Y_{old} \times \sin\theta$
$= 0 \times \cos 90^o - 0 \times \sin 90^o$
$= 0$

Thus, New coordinates of corner A after rotation = (0, 0).

Y_{new}
$= X_{old} \times \sin\theta + Y_{old} \times \cos\theta$
$= 0 \times \sin 90^o + 0 \times \cos 90^o$
$= 0$

For Coordinates B(1, 0)

- Let the new coordinates of corner B after rotation = (X_{new}, Y_{new}).

$$X_{new} = X_{old} \times \cos\theta - Y_{old} \times \sin\theta$$
$$= 1 \times \cos 90^{o} - 0 \times \sin 90^{o}$$
$$= 0$$

- Thus, New coordinates of corner B after rotation = (0, 1).

$$Y_{new} = X_{old} \times \sin\theta + Y_{old} \times \cos\theta$$
$$= 1 \times \sin 90^{o} + 0 \times \cos 90^{o}$$
$$= 1 + 0$$
$$= 1$$

For Coordinates C(1, 1)

Let the new coordinates of corner C after rotation = (X_{new}, Y_{new}).

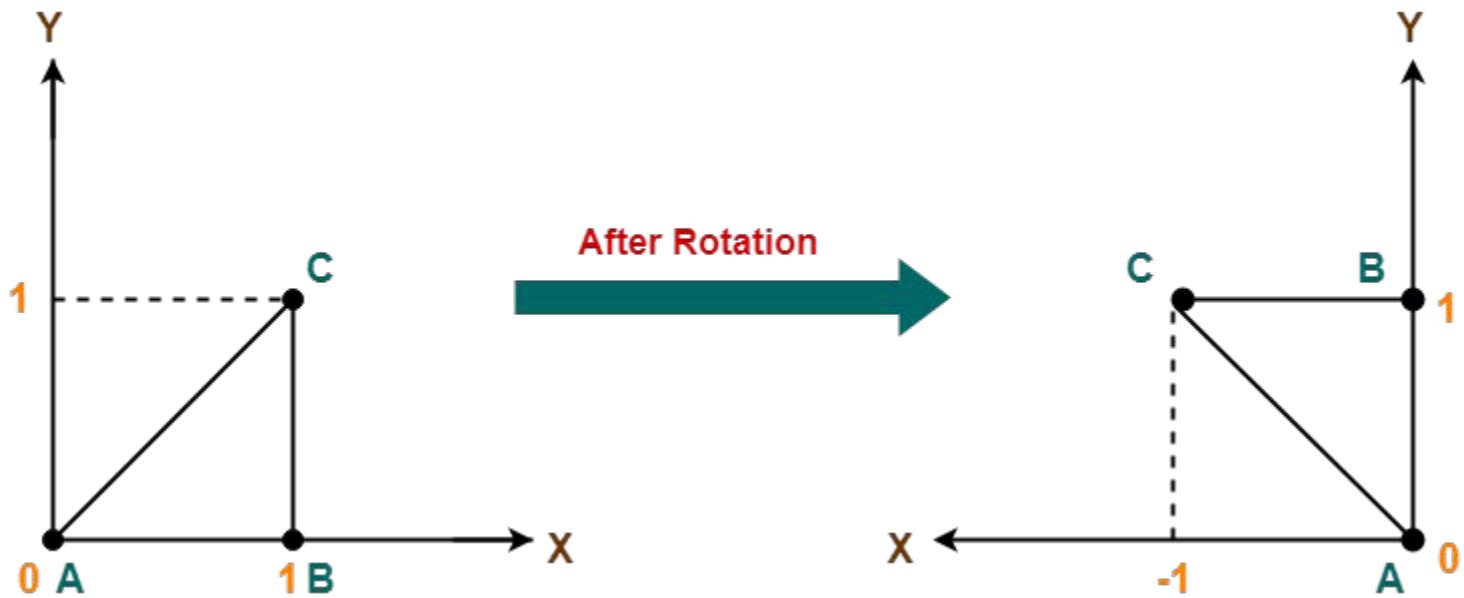
Thus, New coordinates of corner C after rotation = (-1, 1).

Thus, New coordinates of the triangle after rotation = A (0, 0), B(0, 1), C(-1, 1).

X_{new}
$= X_{old} \times \cos\theta - Y_{old} \times \sin\theta$
$= 1 \times \cos 90^o - 1 \times \sin 90^o$
$= 0 - 1$
$= -1$

Y_{new}
$= X_{old} \times \sin\theta + Y_{old} \times \cos\theta$
$= 1 \times \sin 90^o + 1 \times \cos 90^o$
$= 1 + 0$
$= 1$

Scaling

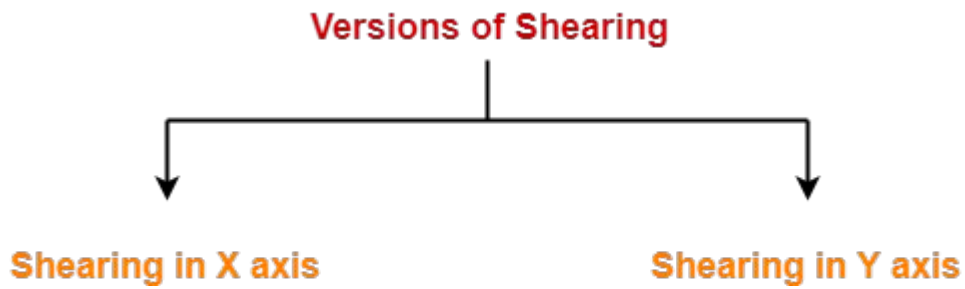


Exercise: Rotate a line with coordinates (0, 0) and (4, 4) by 30^0 in anticlockwise direction.

Shearing

2D Shearing is an ideal technique to change the shape of an existing object in a two dimensional plane.

- In a two dimensional plane, the object size can be changed along X direction as well as Y direction.



A shear in the **x** direction would be as follows:

$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix} = \begin{bmatrix} P_x + hP_y \\ P_y \end{bmatrix} \qquad \begin{bmatrix} Q_x \\ Q_y \end{bmatrix} = \begin{bmatrix} P_x + hP_y \\ P_y + gP_x \end{bmatrix}$$

The quantity h specifies what fraction of the y-coordinate should be added to the x-coordinate, and may be positive or negative

Consider a point object O has to be sheared in a 2D plane.

Let-

- Initial coordinates of the object O = (X_{old}, Y_{old})
- Shearing parameter towards X direction = Sh_x
- Shearing parameter towards Y direction = Sh_y
- New coordinates of the object O after shearing = (X_{new}, Y_{new})

Shearing in X Axis-

Shearing in X axis is achieved by using the following shearing equations-

- $X_{new} = X_{old} + Sh_x \times Y_{old}$
- $Y_{new} = Y_{old}$

Matrix form of the above shearing equations

$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} 1 & Sh_x \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix}$$

Shearing Matrix

(In X axis)

Shearing in Y Axis

- Shearing in Y axis is achieved by using the following shearing equations-

$$X_{new} = X_{old}$$

$$Y_{new} = Y_{old} + Sh_y \times X_{old}$$

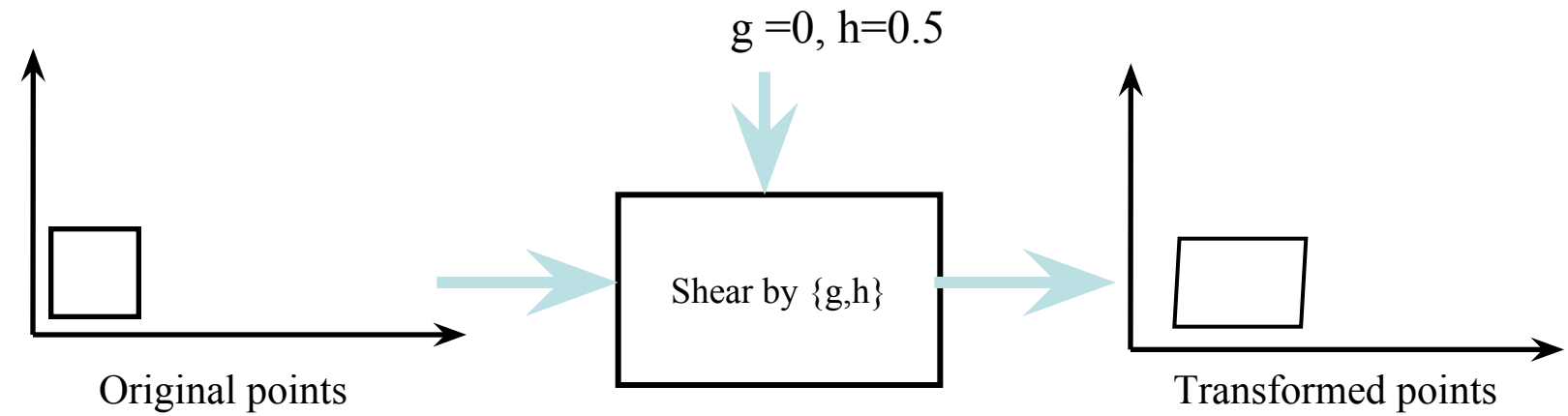
- In Matrix form, the above shearing equations may be represented as-

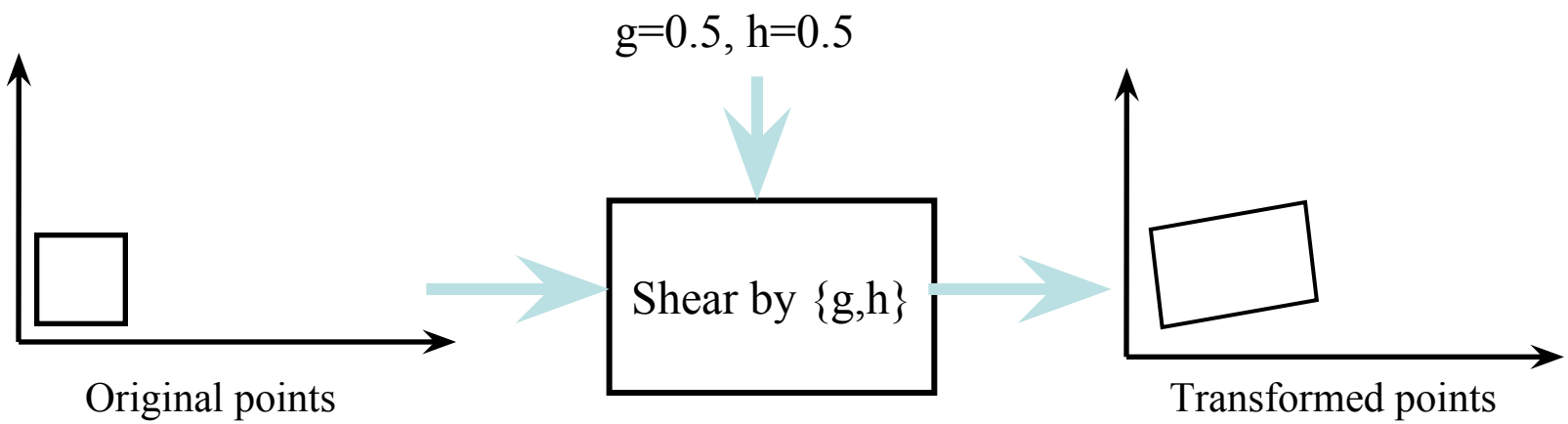
$$\begin{bmatrix} X_{new} \\ Y_{new} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ Sh_y & 1 \end{bmatrix} \times \begin{bmatrix} X_{old} \\ Y_{old} \end{bmatrix}$$

Shearing Matrix

(In Y axis)

Shearing


g =0, h=0.5
Shear by {g,h}
Original points
Transformed points


g=0.5, h=0.5
Shear by {g,h}
Original points
Transformed points

Example: Given a triangle with points (1, 1), (0, 0) and (1, 0). Apply shear parameter 2 on X axis and 2 on Y axis individually and find out the new coordinates of the object.

Solution-

Given-

- Old corner coordinates of the triangle = A (1, 1), B(0, 0), C(1, 0)
- Shearing parameter towards X direction (Sh_x) = 2
- Shearing parameter towards Y direction (Sh_y) = 2

Shearing in X Axis-

For Coordinates A(1, 1)

Let the new coordinates of corner A after shearing = (X_{new}, Y_{new}).

Applying the shearing equations, we have-

- $X_{new} = X_{old} + Sh_x \times Y_{old} = 1 + 2 \times 1 = 3$
- $Y_{new} = Y_{old} = 1$

Thus, New coordinates of corner A after shearing = (3, 1).

For Coordinates B(0, 0)

Let the new coordinates of corner B after shearing = (X_{new}, Y_{new}).
Applying the shearing equations, we have-

- $X_{new} = X_{old} + Sh_x \times Y_{old} = 0 + 2 \times 0 = 0$
- $Y_{new} = Y_{old} = 0$

Thus, New coordinates of corner B after shearing = (0, 0).

For Coordinates C(1, 0)

Let the new coordinates of corner C after shearing = (X_{new}, Y_{new}).

Applying the shearing equations, we have-

- $X_{new} = X_{old} + Sh_x \times Y_{old} = 1 + 2 \times 0 = 1$
- $Y_{new} = Y_{old} = 0$

Thus, New coordinates of corner C after shearing = (1, 0).
Thus, New coordinates of the triangle after shearing in X axis = A (3, 1), B(0, 0), C(1, 0).

Shearing in Y Axis-

For Coordinates A(1, 1)

Let the new coordinates of corner A after shearing = (X_{new}, Y_{new}).

Applying the shearing equations, we have-

- $X_{new} = X_{old} = 1$
- $Y_{new} = Y_{old} + Sh_y \times X_{old} = 1 + 2 \times 1 = 3$

Thus, New coordinates of corner A after shearing = (1, 3).

For Coordinates B(0, 0)

Let the new coordinates of corner B after shearing = (X_{new}, Y_{new}).

Applying the shearing equations, we have-

- $X_{new} = X_{old} = 0$
- $Y_{new} = Y_{old} + Sh_y \times X_{old} = 0 + 2 \times 0 = 0$

Thus, New coordinates of corner B after shearing = (0, 0).

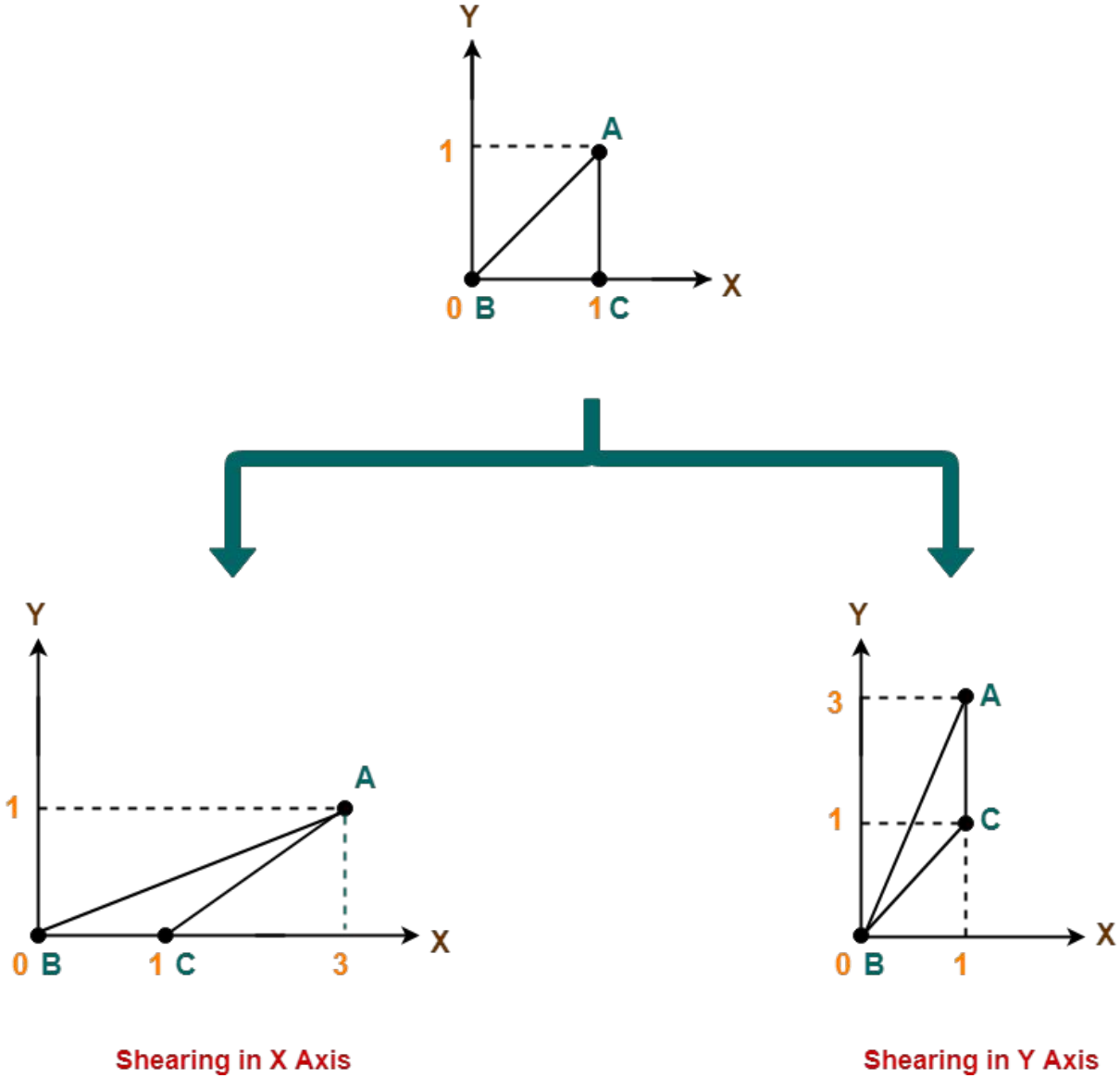
For Coordinates C(1, 0)

Let the new coordinates of corner C after shearing = (X_{new}, Y_{new}).

Applying the shearing equations, we have-

- $X_{new} = X_{old} = 1$
- $Y_{new} = Y_{old} + Sh_y \times X_{old} = 0 + 2 \times 1 = 2$

Thus, New coordinates of corner C after shearing = (1, 2).
Thus, New coordinates of the triangle after shearing in Y axis = A (1, 3), B(0, 0), C(1, 2).



Shearing in X Axis

Shearing in Y Axis

Matrix Representation

All affine transformations in 2D can be generically described in terms of a generic equation as follows:

$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} P_x \\ P_y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

i.e.

$$\mathbf{Q} = \mathbf{MP} + \mathbf{t}$$

Transformations Summary

- 1: Translation

$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} P_x \\ P_y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

- 2: Scaling

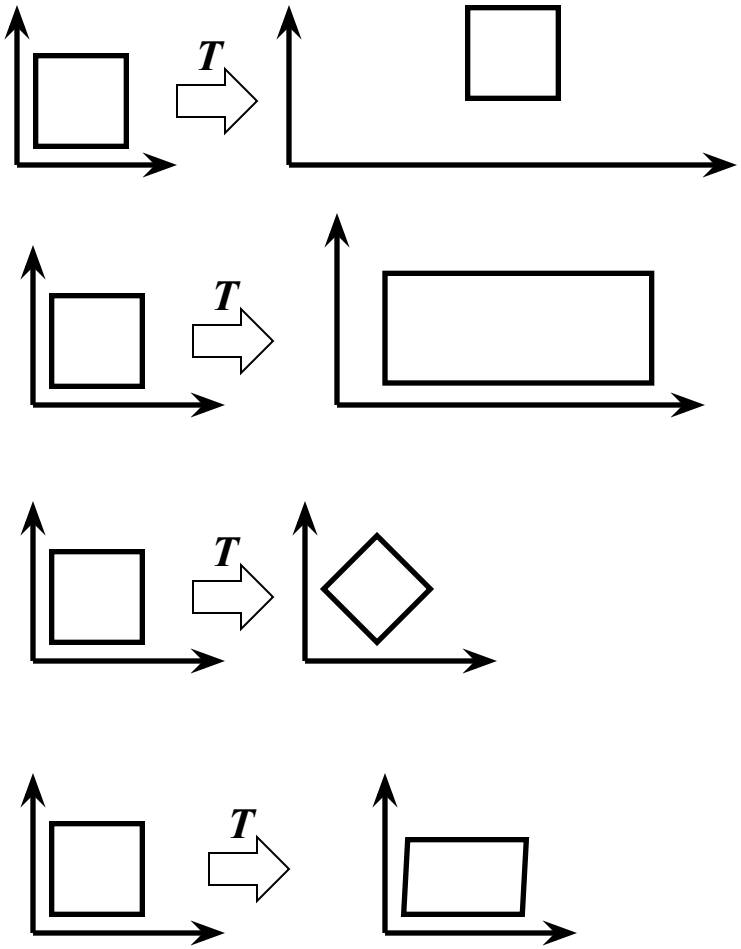
$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} P_x \\ P_y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- 3: Rotation

$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} P_x \\ P_y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- 4: Shearing

$$\begin{bmatrix} Q_x \\ Q_y \end{bmatrix} = \begin{bmatrix} 1 & h \\ g & 1 \end{bmatrix} \begin{bmatrix} P_x \\ P_y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



Problem

An affine transformation is composed of a linear combination followed by a translation

Unfortunately, the translation portion is not a matrix multiplication but must instead be added as an extra term, or vector – this is inconvenient

Homogeneous coordinates

- One of the many purposes of using homogeneous coordinates is to capture the concept of infinity. In the Euclidean coordinate system, infinity is something that does not exist. Many geometric concepts and computations can be greatly simplified if the concept of infinity is used. This will become very clear when we move to curves and surfaces design. Without the use of homogeneous coordinates system, it would be difficult to design certain classes of very useful curves and surfaces in computer graphics and computer-aided design.
- Let us consider two real numbers, a and h, and compute the value of a/h. Let us hold the value of a fixed and vary the value of h. As h gets smaller, the value of a/h gets larger. If h approaches zero, a/h approaches infinity! Thus, to capture the concept of infinity, we use two numbers a and h to represent a value v, $v=a/h$. If h is not zero, the value is exactly a/h. Otherwise, we identify the infinite value with $(a,0)$. Therefore, the concept of infinity can be represented with a number pair like (a, h) or as a quotient a/h.
- Let us apply this to the xy-coordinate plane. If we replace x and y with x/h and y/h, a function $f(x,y)=0$ becomes $f(x/h,y/h)=0$.

Given a point (x,y,h) in homogeneous coordinates, what is its corresponding point in the xy-plane?
The answer must be (x/h,y/h).

Thus, a point (3,4,5) in homogeneous coordinates converts to point (3/5,4/5)=(0.6,0.8) in the xy-plane.
Similarly, a point (x,y,z,h) in homogeneous coordinates converts to a point (x/h,y/h,z/h) in space.
Conversely, what is the homogeneous coordinates of a point (x,y) in the xy-plane?
Answer: (x,y,1)! That is, let the h component be 1.

In fact, this is only part of the story, because the answer is ***not*** unique.
The homogeneous coordinates of a point (x,y) in the xy-plane is (xh, yh, h) for any non-zero h.
Why is this true? Because (xh, yh, h) is converted back to (x,y).

Converting from a homogeneous coordinates to a conventional one is <u>unique</u>; but, converting a conventional coordinates to a homogeneous one is <u>not</u>.
For example, a point (4,2,3) in space is convert to (4h, 2h, 3h, h) for any non-zero h

Homogeneous Coordinates

A point (x, y) can be re-written in **homogeneous coordinates** as (x_h, y_h, h)

The **homogeneous parameter** h is a non-zero value such that:

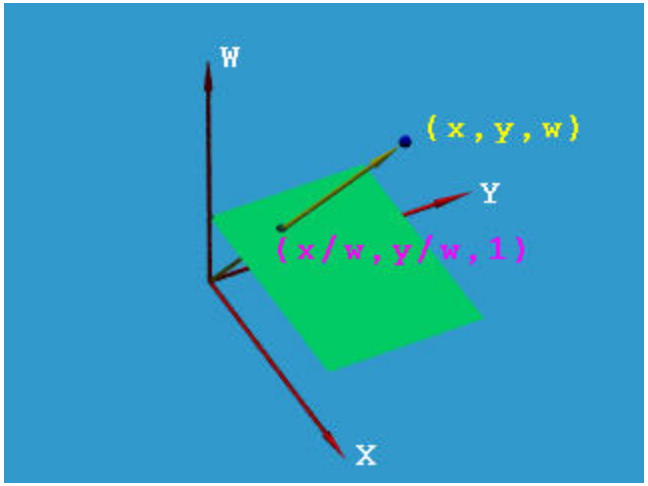
$$x = \frac{x_h}{h} \qquad y = \frac{y_h}{h}$$

We can then write any point (x, y) as $(h.x, h.y, h)$

We can conveniently choose $h = 1$ so that (x, y) becomes $(x, y, 1)$

A Simple Geometric Interpretation

Given a homogeneous coordinate (x,y,w) of a point in the xy-plane, let us consider (x,y,w) to be a point in space whose coordinate values are x, y and w for the x-, y- and w- axes, respectively. The line joining this point and the coordinate origin intersects the plane $w = 1$ at a point $(x/w, y/w, 1)$. The figure illustrates this concept.
here, h=w.



This transformation treats a two-dimensional homogeneous point as a point in three-dimensional space and projects (from the coordinate origin) this three-dimensional point to the plane w=1. Therefore, as a homogeneous point moves on a curve defined by homogeneous polynomial $f(x,y,w)$=0, its corresponding point moves in three-dimensional space, which, in turn, is projected to the plane w=1. Of course, $(x/w,y/w)$ moves on a curve in plane w=1.
While the conversion from the conventional Euclidean coordinates to homogeneous coordinates is unique, the opposite direction is not because all points on the line joining the origin and (x,y,w) will be projected to $(x/w,y/w,1)$. This is also an important concept to be used in later lectures.

- The use of homogeneous coordinate system is vital when there are multiple operation which include translation, as in this coordinate system; translation is also represented as multiplication.
- Consider any point P(x y) which can be expressed as

$$(x, y) \rightarrow (x_h, y_h, h)$$

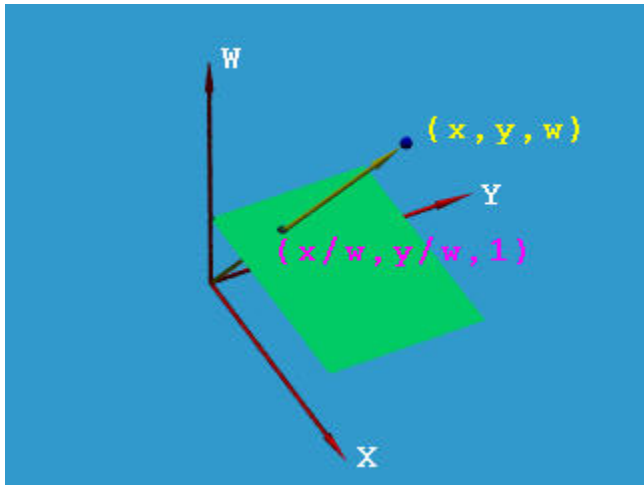
$$x = \frac{x_h}{h}$$

$$y = \frac{y_h}{h}$$

Matrix representations

$$\text{Translation} = \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous		Cartesian		
$(1,2,3)$	$\Rightarrow$	$\left(\frac{1}{3}, \frac{2}{3}\right)$		
$(2,4,6)$	$\Rightarrow$	$\left(\frac{2}{6}, \frac{4}{6}\right)$	$=$	$\left(\frac{1}{3}, \frac{2}{3}\right)$
$(4,8,12)$	$\Rightarrow$	$\left(\frac{4}{12}, \frac{8}{12}\right)$	$=$	$\left(\frac{1}{3}, \frac{2}{3}\right)$
$\vdots$		$\vdots$		
$(1a, 2a, 3a)$	$\Rightarrow$	$\left(\frac{1a}{3a}, \frac{2a}{3a}\right)$	$=$	$\left(\frac{1}{3}, \frac{2}{3}\right)$



Cartesian and homogeneous coordinates

$(X_a, Y_a, Z_a) = (X_p / W_p, Y_p / W_p, Z_p / W_p)$

Cartesian	Homogeneous
(4, 2, 5)	(4, 2, 5, 1)
(40, 20, 50)	(4, 2, 5, 0.1)
(400, 200, 500)	(4, 2, 5, 0.01)
(4000, 2000, 5000)	(4, 2, 5, 0.001)
(?, ?, ?)	(4, 2, 5, 0)

Should be somewhere on the same line, but further than any other point, right?

Why Homogeneous Coordinates?

Mathematicians commonly use homogeneous coordinates as they allow scaling factors to be removed from equations

We will see in a moment that all of the transformations we discussed previously can be represented as 3*3 matrices

Using homogeneous coordinates allows us use matrix multiplication to calculate transformations – extremely efficient!

Homogeneous Coordinates

The "trick" we use is to add an additional component 1 to both P and Q, and also a third row and column to M, consisting of zeros and a 1

i.e.

$$\begin{bmatrix} Q_x \\ Q_y \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} P_x \\ P_y \\ 1 \end{bmatrix}$$

Translation by {tx, ty}

$$M = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

Rotate by θ:

$$M = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Scale by Sx, Sy

$$M = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Shear by g, h:

$$M = \begin{bmatrix} 1 & h & 0 \\ g & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Homogeneous Translation

The translation of a point by *(dx, dy)* can be written in matrix form as:

$$\begin{bmatrix} 1 & 0 & dx \\ 0 & 1 & dy \\ 0 & 0 & 1 \end{bmatrix}$$

Representing the point as a homogeneous column vector we perform the calculation as:

$$\begin{bmatrix} 1 & 0 & dx \\ 0 & 1 & dy \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 1*x+0*y+dx*1 \\ 0*x+1*y+dy*1 \\ 0*x+0*y+1*1 \end{bmatrix} = \begin{bmatrix} x+dx \\ y+dy \\ 1 \end{bmatrix}$$

Remember Matrix Multiplication

Recall how matrix multiplication takes place:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a*x+b*y+c*z \\ d*x+e*y+f*z \\ g*x+h*y+i*z \end{bmatrix}$$

Homogenous Coordinates

To make operations easier, 2-D points are written as homogenous coordinate column vectors

Translation: $$\begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x+tx \\ y+ty \\ 1 \end{bmatrix} : v' = T(tx, ty)v$$

Scaling: $$\begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} s_x \times x \\ s_y \times y \\ 1 \end{bmatrix} : v' = S(s_x, s_y)v$$

Homogenous Coordinates (cont…)

Rotation:

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta \times x - \sin\theta \times y \\ \sin\theta \times x + \cos\theta \times y \\ 1 \end{bmatrix} : v' = R(\theta)v$$

Inverse Transformations

Transformations can easily be reversed using inverse transformations

$$T^{-1} = \begin{bmatrix} 1 & 0 & -dx \\ 0 & 1 & -dy \\ 0 & 0 & 1 \end{bmatrix}$$

$$R^{-1} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S^{-1} = \begin{bmatrix} \frac{1}{s_x} & 0 & 0 \\ 0 & \frac{1}{s_y} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Composite Transformations

Translations

If 2 successive translation vectors (dx_1, dy_1) and (dx_2, dy_2) are applied to a coordinate position V, the final transformed location $V^{/}$ is calculated as

$$V^{/} = \begin{matrix} T(dx_2, dy_2).\{T(dx_1, dy_1).V\} \\ T(dx_2, dy_2).T(dx_1, dy_1).V \end{matrix}$$

Where P and $P^{/}$ are represented as homogeneous coordinate column vectors.

$$\begin{bmatrix} 1 & 0 & dx_2 \\ 0 & 1 & dy_2 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & dx_1 \\ 0 & 1 & dy_1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & dx_1 + dx_2 \\ 0 & 1 & dy_1 + dy_2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$or \qquad T(dx_2, dy_2).T(dx_1, dy_1) = T(dx_1 + dx_2, dx_2 + dy_2)$$

Which demonstrates that 2 successive translations are additive

Composite Transformations

Rotations

If 2 successive rotation vectors are applied to a coordinate position V, the final rotated location V' is calculated as

$$V' = \begin{matrix} R(\theta_2).\{R(\theta_1).V\} \\ \{R(\theta_2).R(\theta_1)\}.V \end{matrix}$$

By multiplying the 2 rotation matrices, we can verify that 2 scuccessive rotations are additive:.

$$R(\theta_2).R(\theta_1) = R(\theta_1 + \theta_2)$$

So that the final rotated coordinates can be calculated with the composite matrix as

$$V' = R(\theta_1 + \theta_2).V$$

Composite Transformations

Scalings

Concatenating transformation matrices for 2 successive scaling operations produces the following composite scaling matrix:

$$\begin{bmatrix} S_{x2} & 0 & 0 \\ 0 & S_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} S_{x1} & 0 & 0 \\ 0 & S_{y1} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} S_{x1}.S_{x2} & 0 & 0 \\ 0 & S_{y1}.S_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$or \qquad S(S_{x2}, S_{y_2}).S(S_{x1}, S_{y_1}) = S(S_{x1}.S_{x2}, S_{y_1}.S_{y_2})$$

Which demonstrates that 2 successive scalings are multiplicative.

Thus if we were to triple the size of an object twice in succession, the final size would be nine times that of the original.

Combining Transformations

A number of transformations can be combined into one matrix to make things easy

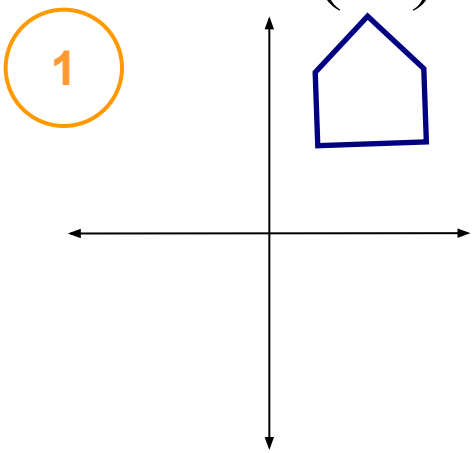
- Allowed by the fact that we use homogenous coordinates

Imagine rotating a polygon around a point other than the origin

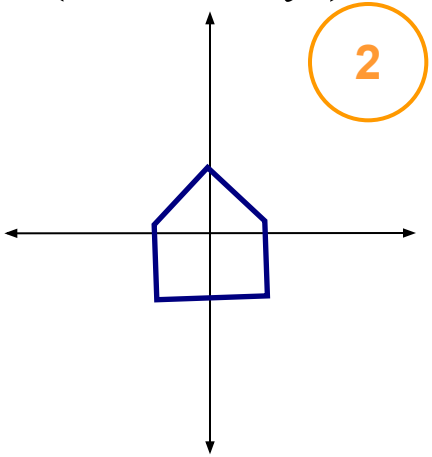
- Transform to centre point to origin
- Rotate around origin
- Transform back to centre point

Combining Transformations (cont…)

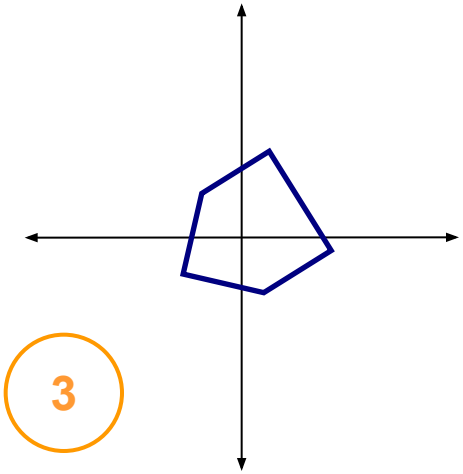
$House\,(H)$



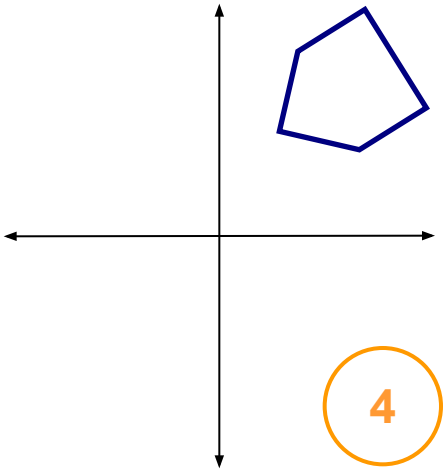
$T(-dx,-dy)H$



$R(\theta)T(-dx,-dy)H$



$T(dx,dy)R(\theta)T(-dx,-dy)H$



Combining Transformations (cont…)

The three transformation matrices are combined as follows

$$\begin{bmatrix} 1 & 0 & dx \\ 0 & 1 & dy \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -dx \\ 0 & 1 & -dy \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$v' = T(dx, dy)R(\theta)T(-dx, -dy)v$$

REMEMBER: Matrix multiplication is not commutative so order matters

Summary

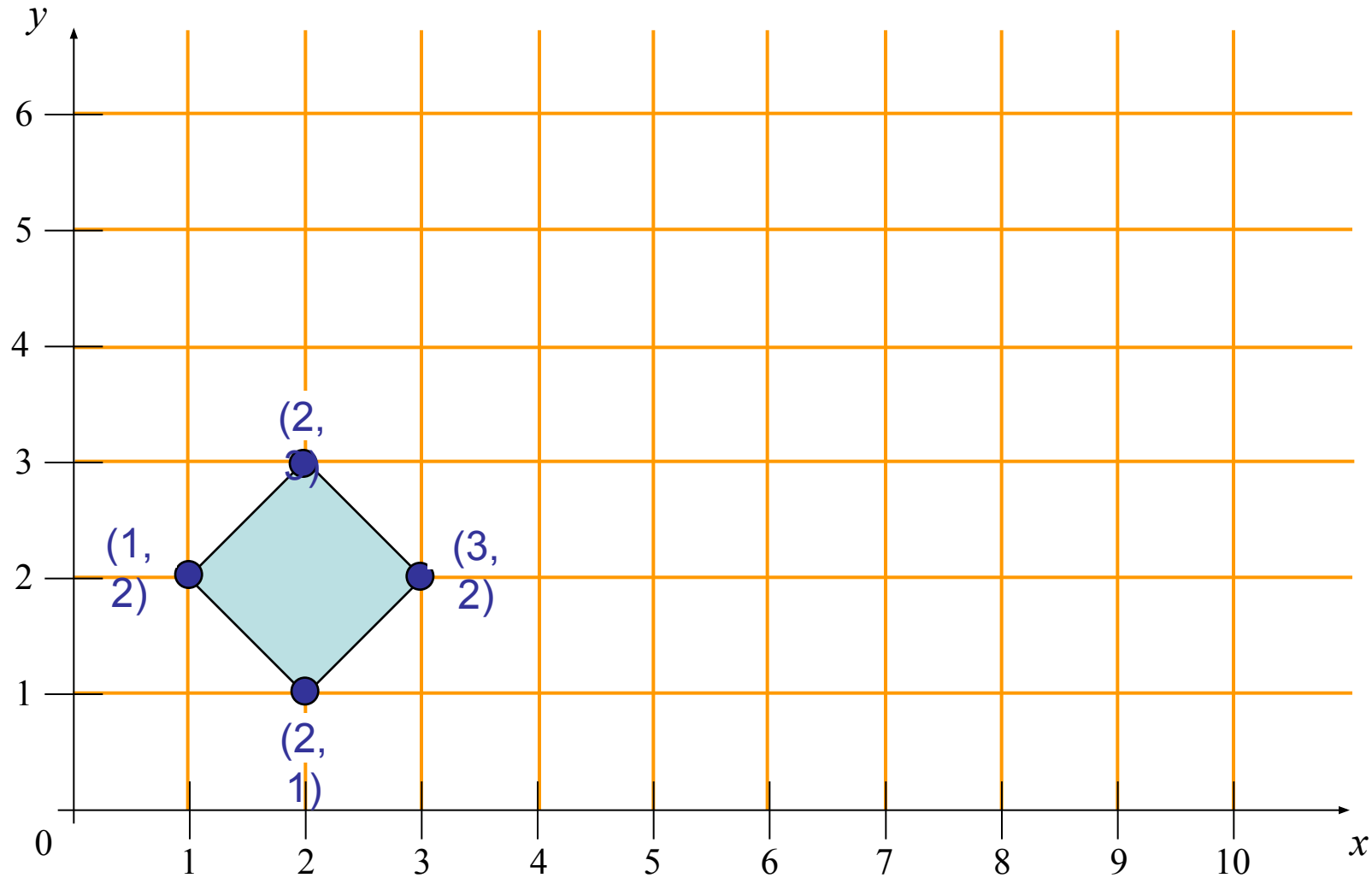
Learnings:

- 2D Transformations
  - Translation
  - Scaling
  - Rotation
- Homogeneous coordinates
- Matrix multiplications
- Combining transformations

Next time we'll start to look at how we take these abstract shapes etc and get them on-screen

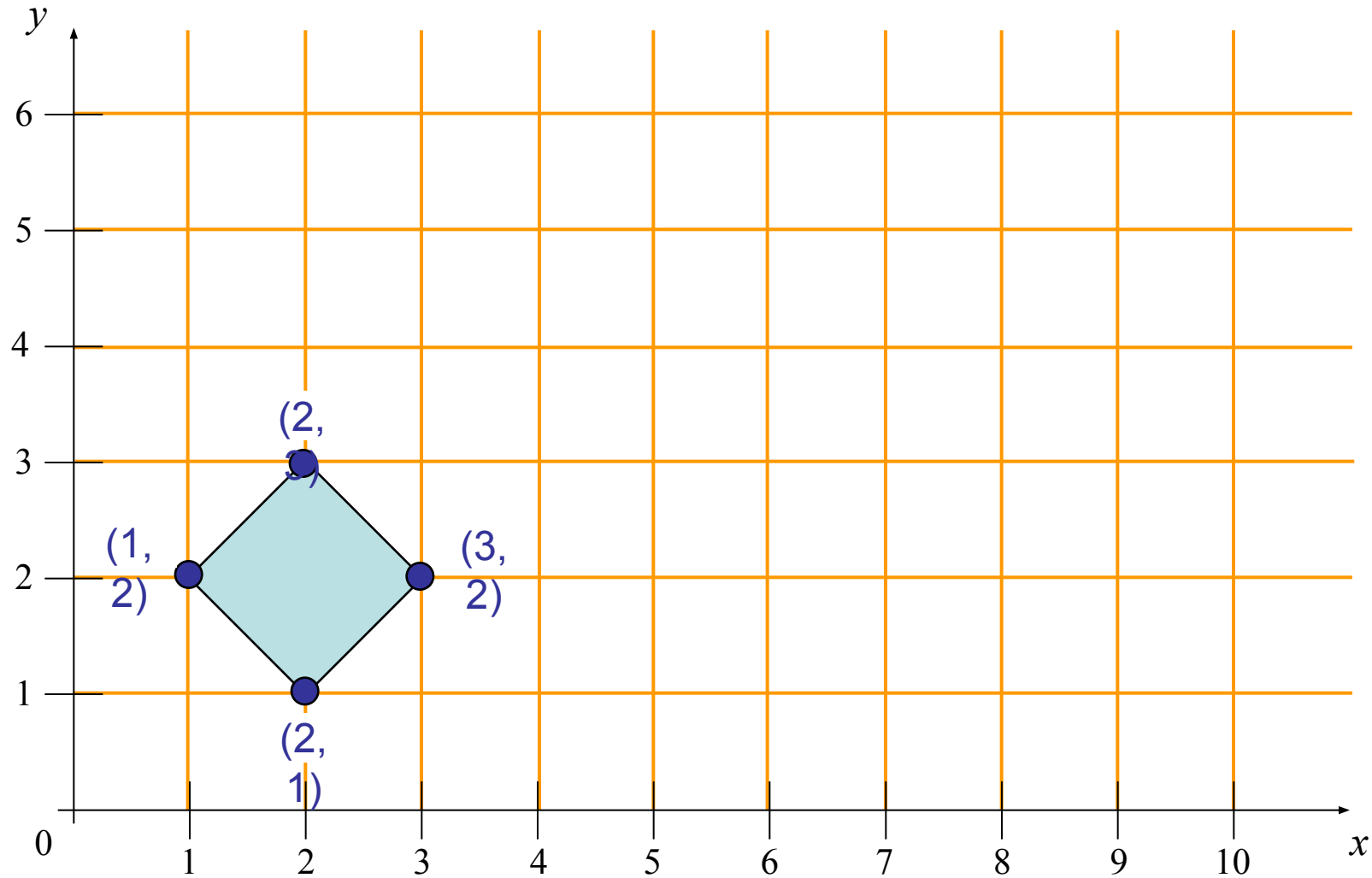
Exercises 1

Translate the shape below by (7, 2)



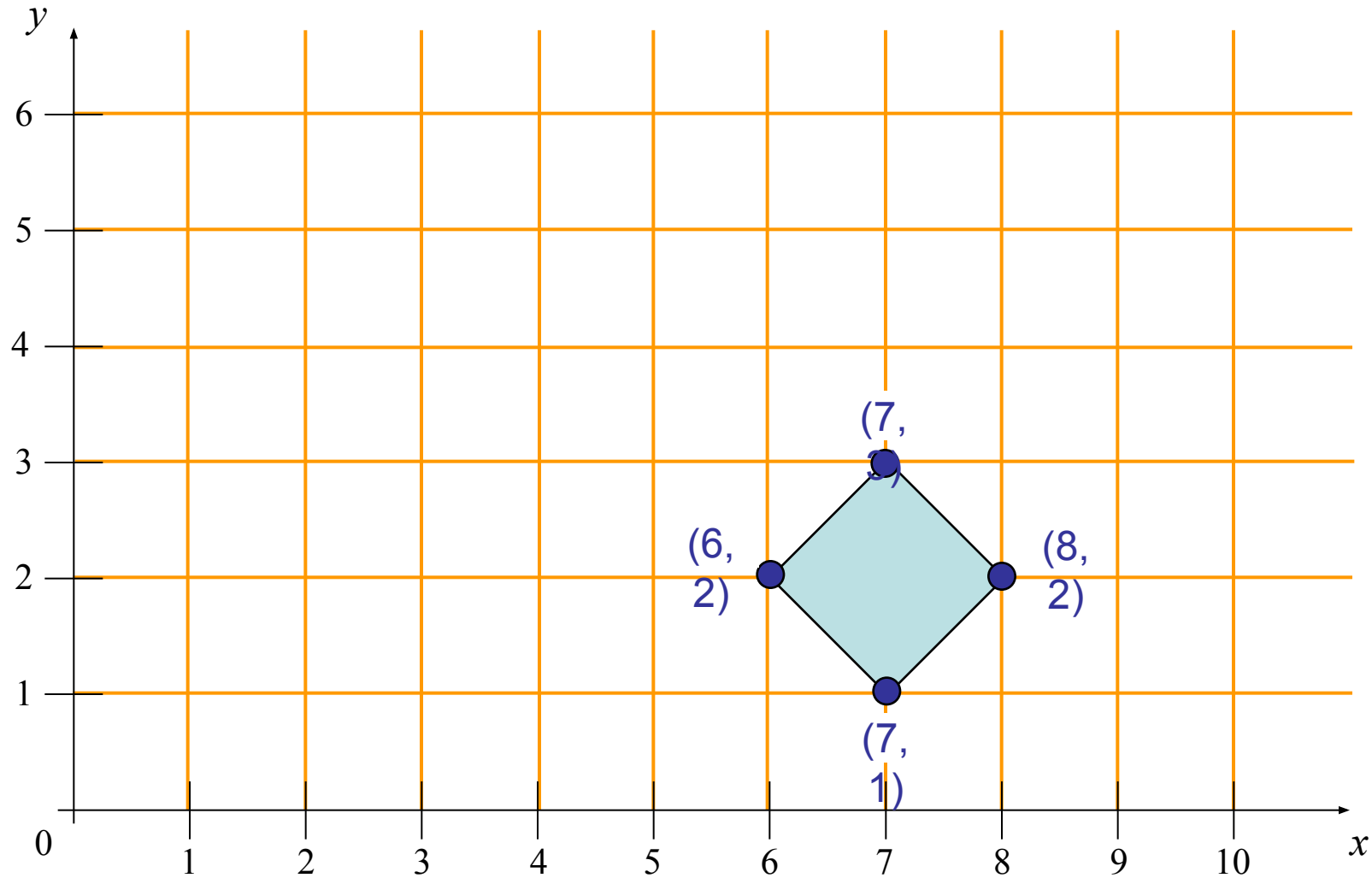
Exercises 2

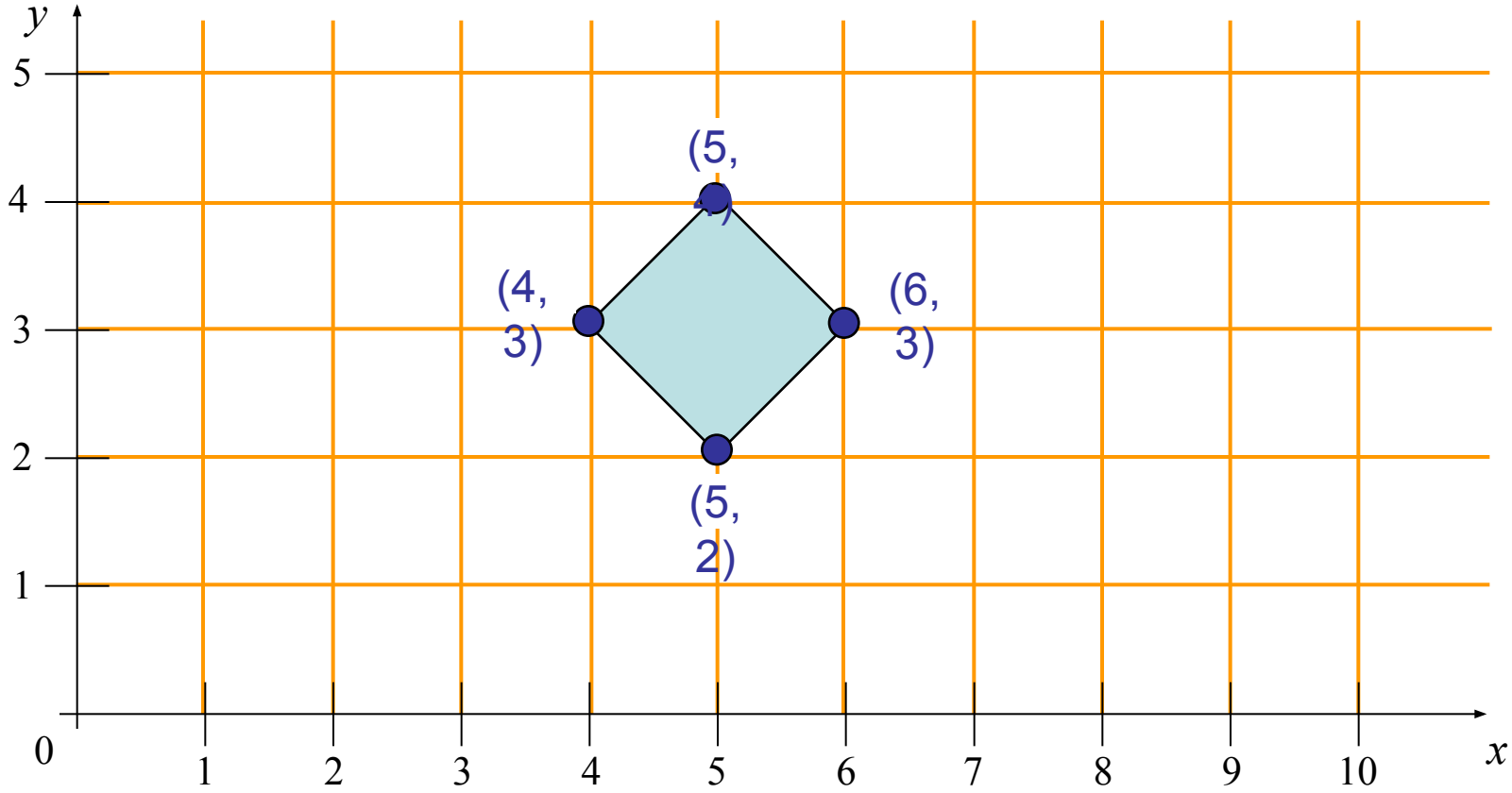
Scale the shape below by 3 in x and 2 in y



Exercises 3

Rotate the shape below by 30° about the origin



Exercises 5

Using matrix multiplication calculate the rotation of the shape below by 45° about its centre (5, 3)